WHO GETS RECOGNIZED, AND WHEN?

A Data Study of MoMA's Acquisition Patterns

Data Science Bootcamp — Capstone Project

160,699 artworks · BigQuery + dbt + Python · Wikidata enrichment

Research Question and Data

Who does a modern art museum collect, and when?

Date
Year the work was made

DateAcquired
Year it entered the collection

acquisition lag = DateAcquired – Date

This project asks which groups of artists and movements MoMA recognized faster, and which later; and whether that difference is real, or just a chronological illusion.

DATA

160,699 artworks · 15,932 artists — MoMA GitHub dataset + Wikidata SPARQL enrichment (movement, geography). Layered BigQuery + dbt architecture, analysis in Python (Colab).

Overview

Before the hypotheses: what does the overall picture show?

160,699

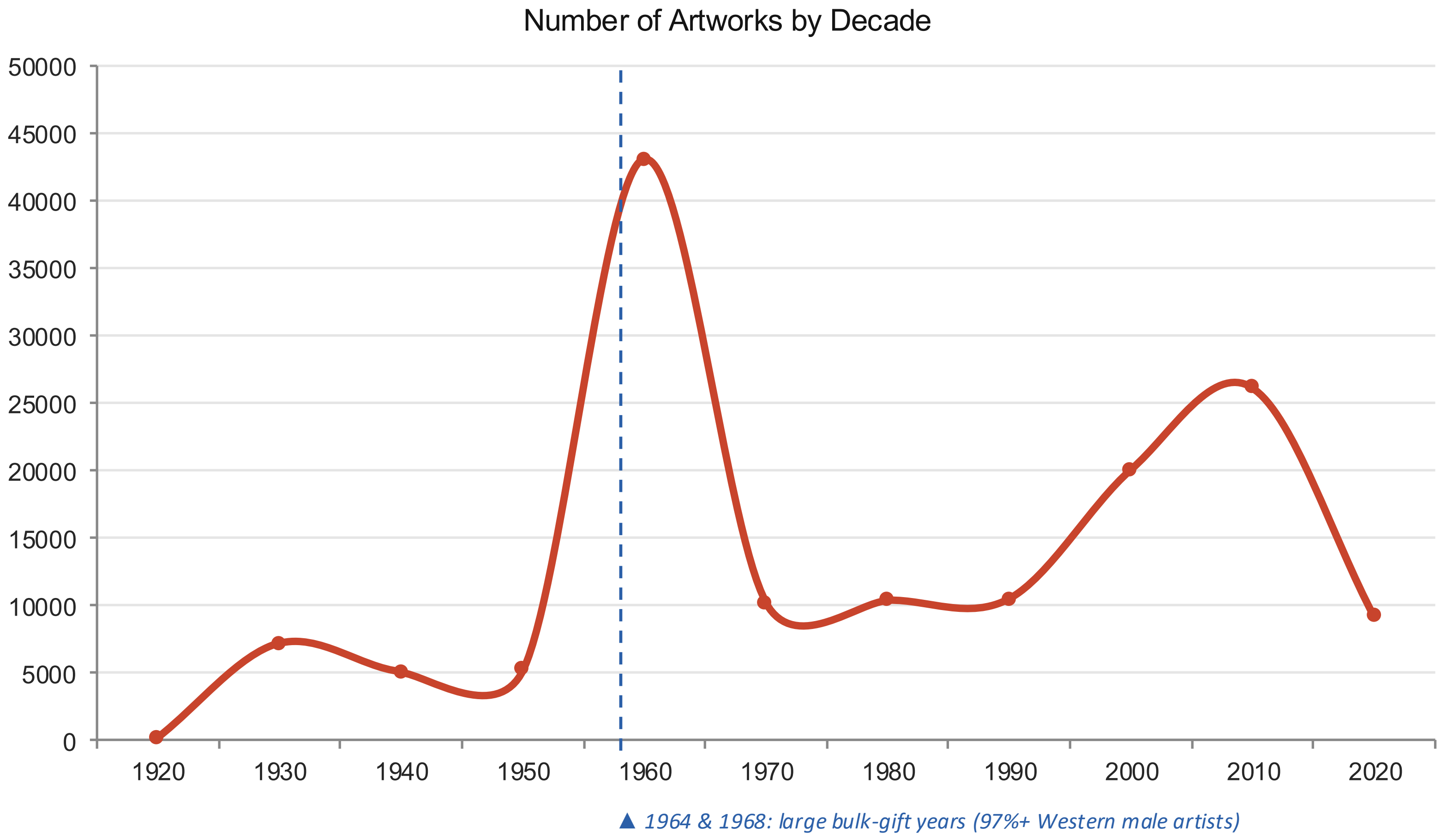
Total artworks

15,932

Total artists

21 years

Median acquisition lag



So what does this general behavior depend on?

We tested four hypotheses — each follows the same framework: question → raw finding → statistical validation

H1

Gender

Does acquisition lag differ between female and male artists?

H2

Geography

Is the gap between Western and non-Western artists permanent or temporary?

H3

Art Movement

Which movements did MoMA recognize early, and which late?

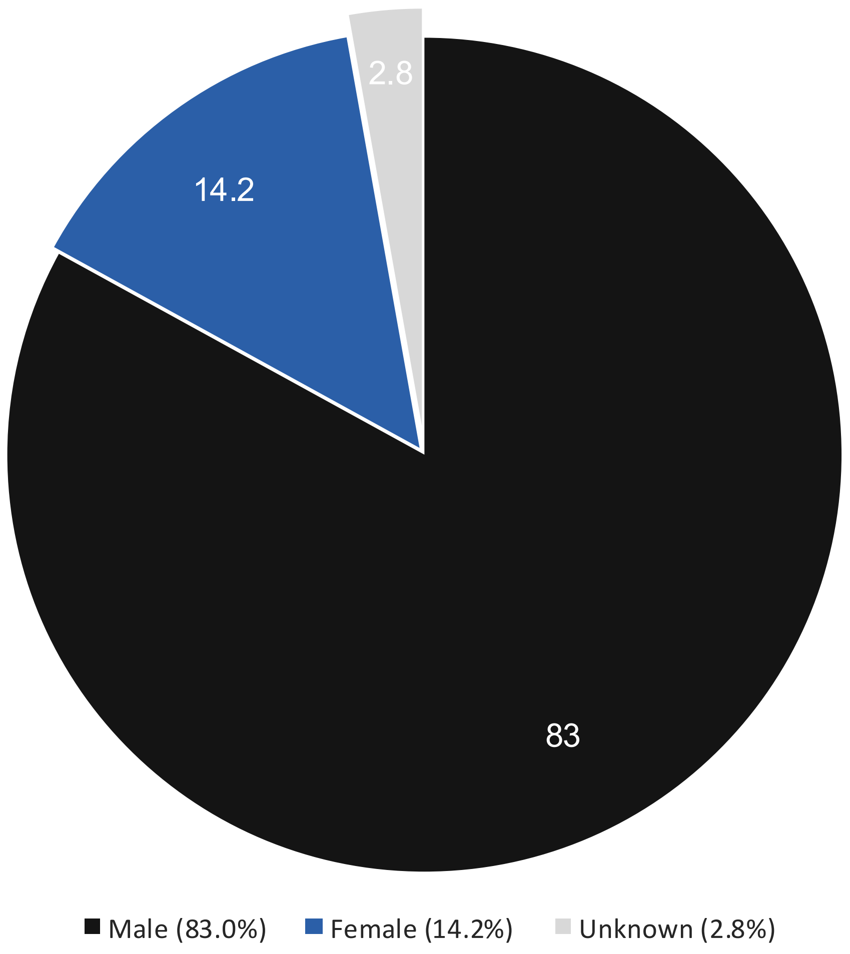
H4

Temporal Consistency

Is MoMA's behavior predictable, or random?

H1 · Gender — Overall Distribution

Gender composition of the collection



General observation

The vast majority of the collection (83%) is by male artists. This is the overall distribution — the real question is how this ratio has changed over time, and whether acquisition behavior differs by gender.

Next question

Does acquisition lag differ by gender? And is that difference a real behavioral gap, or simply a result of works by female artists being more recent on average?

H1 · Gender — From Raw Finding to Verified Result

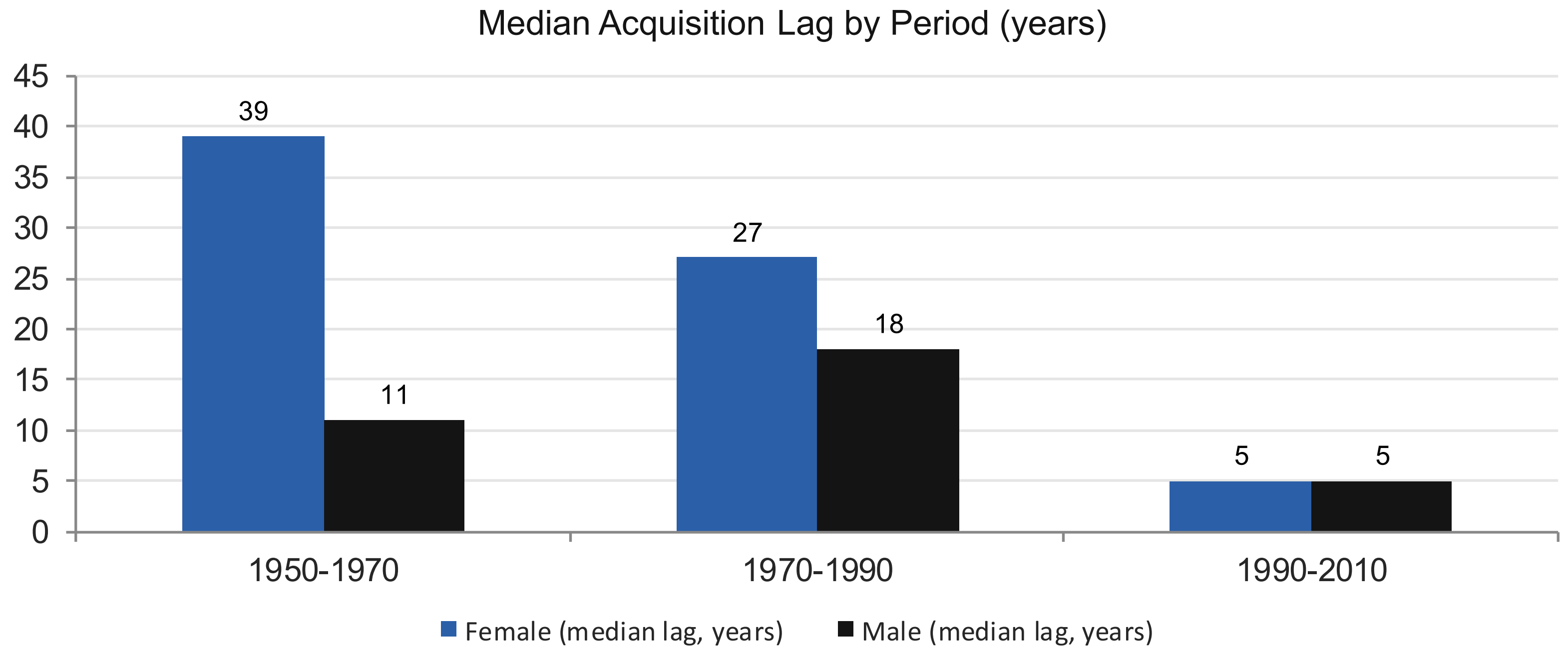
Controlling for a confounding variable reversed the picture

RAW FINDING (uncontrolled)

Works by female artists appear to be acquired FASTER — but this is misleading: they're already more recent on average.

CONTROLLED FINDING (period-matched)

Comparing works made in the same period reveals the real picture: works by female artists were historically acquired SLOWER.



Mann-Whitney U Test

1950-70

$p < 0.001$

Significant

1970-90

$p < 0.001$

Significant

1990-2010

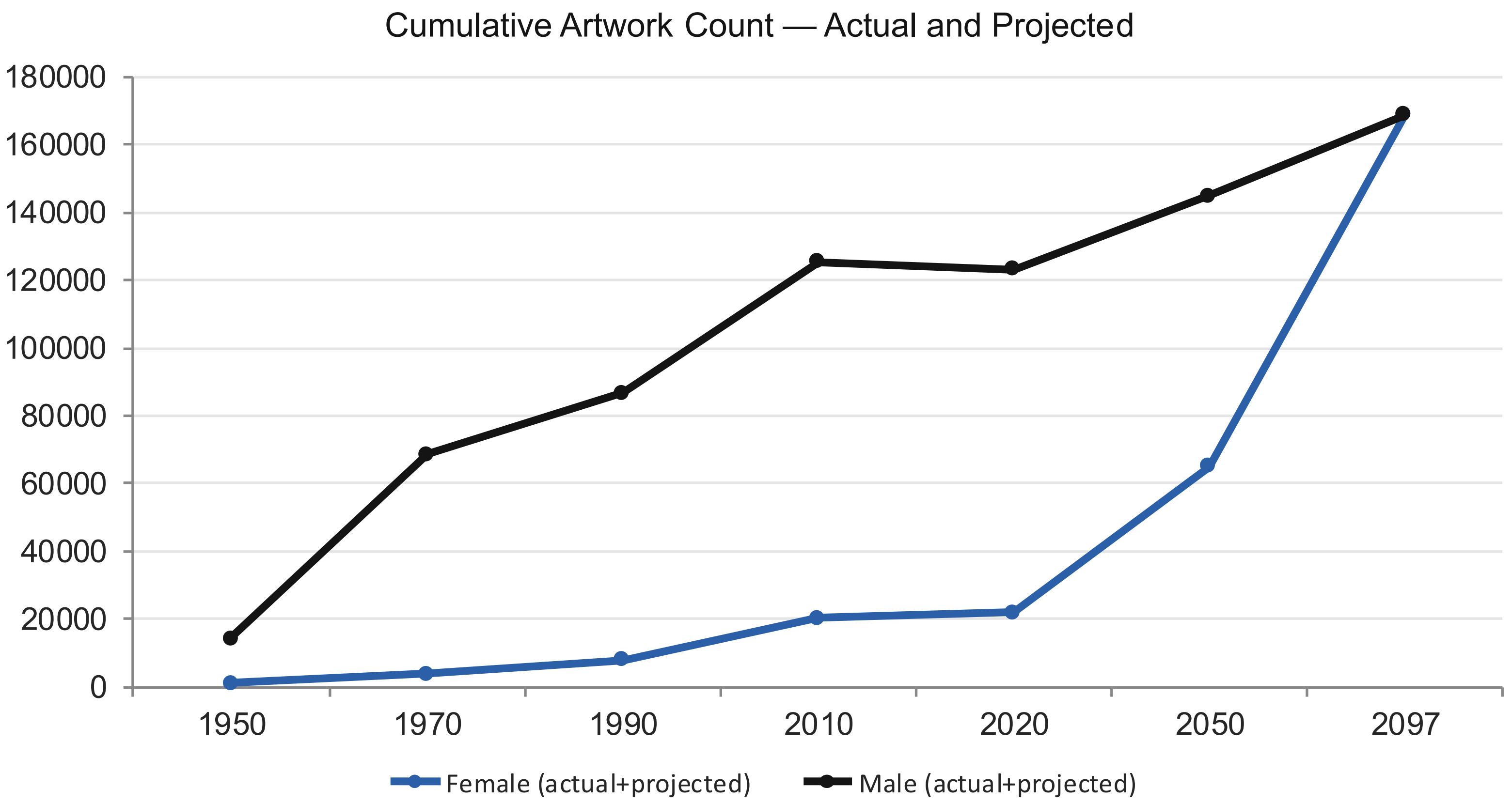
$p = 0.418$

Not significant

Robustness check: excluding the 1960 bulk-gift year from the 1950-70 calculation shrank the gap (28→25 years), but it held.

H1 · Gender — Artwork Count Projection

The lag closed, but representation is still unbalanced



2097

Estimated crossover year

As of 2020:

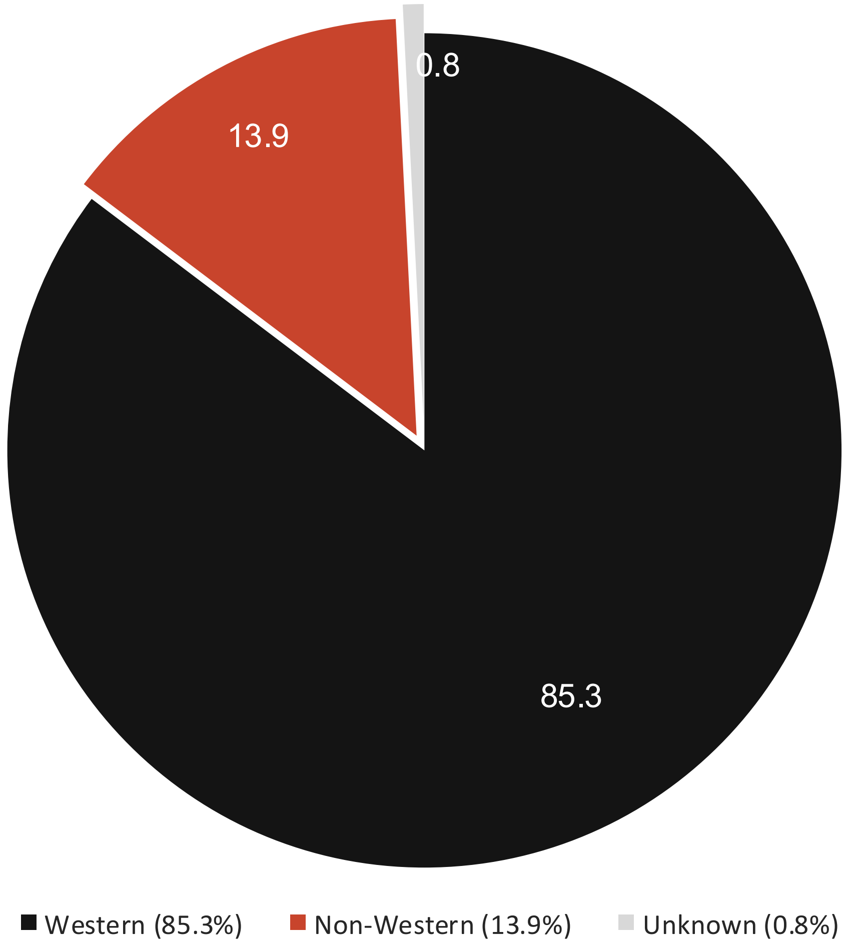
21,828 works by female artists
123,378 works by male artists

Method: a compound (exponential) growth model based on period growth rates — a year-by-year crossover simulation.

Conclusion: The lag behavior corrected itself (H1), but numerical parity is expected to require ~80 more years at the currenttrend.

H2 · Geography — Overall Distribution

Geographic composition of the collection



General observation

The vast majority of the collection (85.3%) is by Western artists. Unlike H1, the production-year gap here is small (~9 years) — suggesting the lag gap can't be explained by chronology.

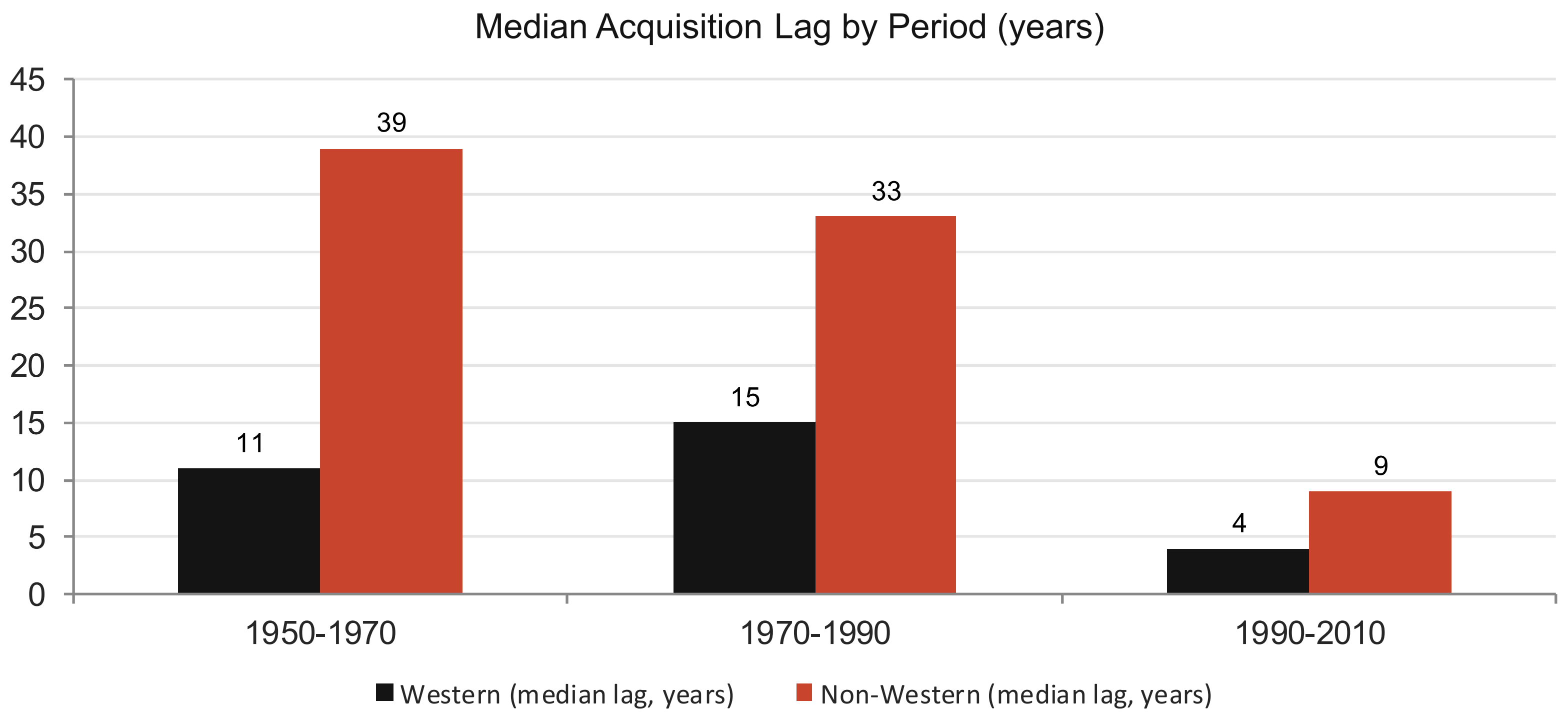
Next question

Is the lag gap between Western and non-Western artists permanent or temporary? Does it close over time like H1, or does it behave differently?

H2 · Geography — From Raw Finding to Verified Result

A persistent pattern between Western and non-Western artists

UNLIKE H1: the production-year gap is small (~9 years) — the lag gap can't be explained by chronology.



Mann-Whitney U Test

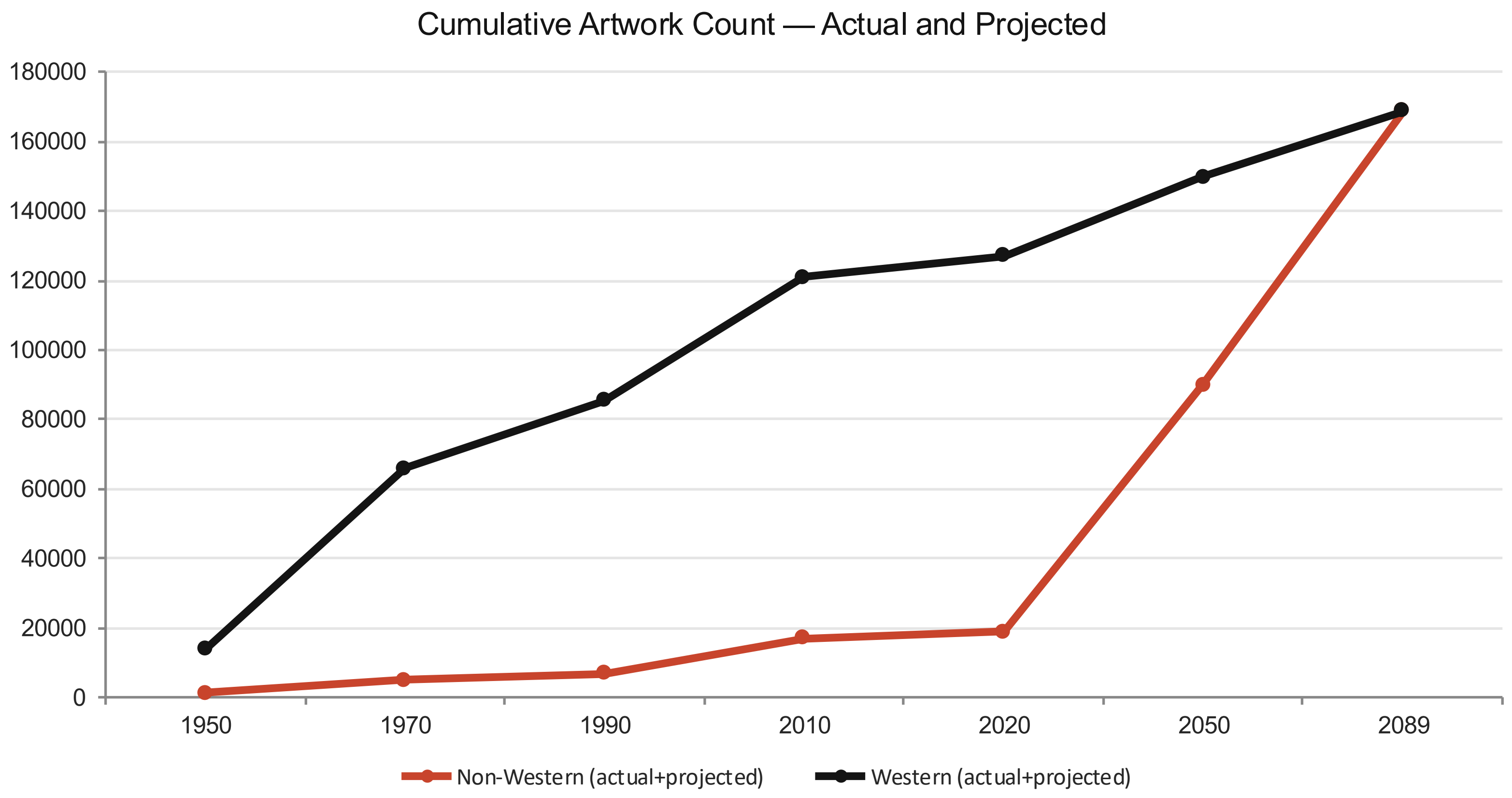
1950-70
$p < 0.001$
Significant

1970-90
$p < 0.001$
Significant

1990-2010
$p < 0.001$
Significant

H2 · Geography — Projection

Two inequalities, two different dynamics



~2089

Estimated crossover year
Non-Western ~71% growth, Western stagnant

As of 2020:

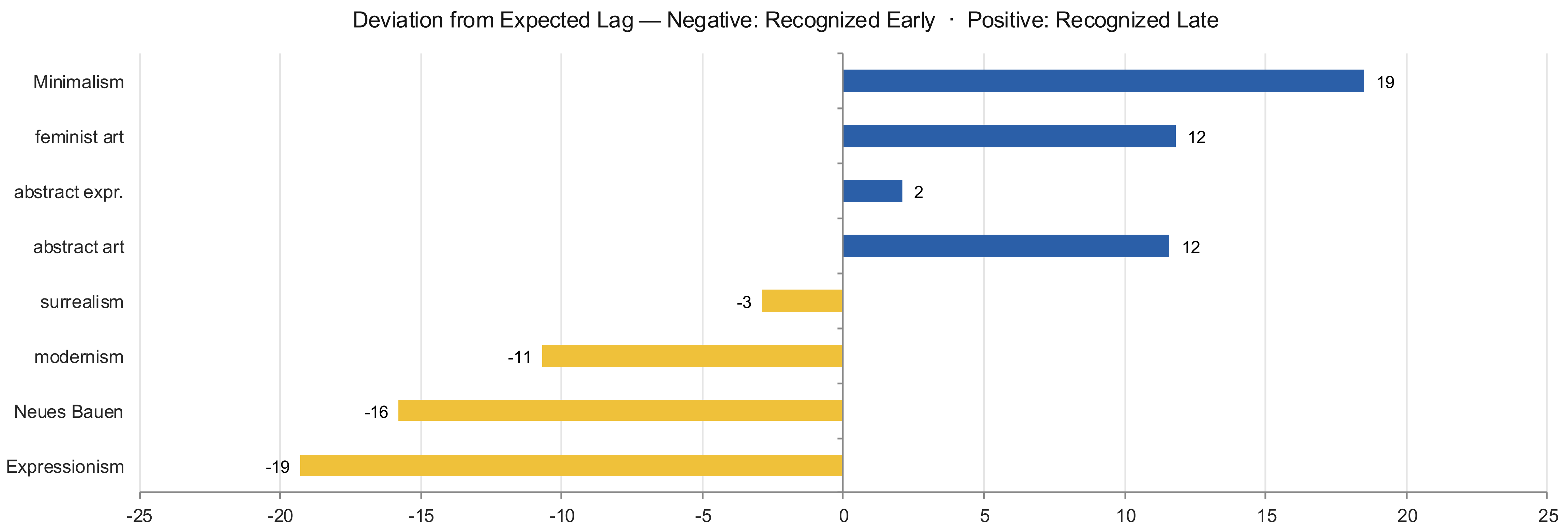
18,969 non-Western works
127,205 Western works

Method: a compound (exponential) growth model based on period growth rates — a year-by-year crossover simulation.

Conclusion: The gap in lag behavior is narrowing but not closing. Numerical parity is expected to require ~70 more years at the current trend.

H3 · Art Movement — Early and Late Recognition

Which movements did MoMA recognize early, and which late?

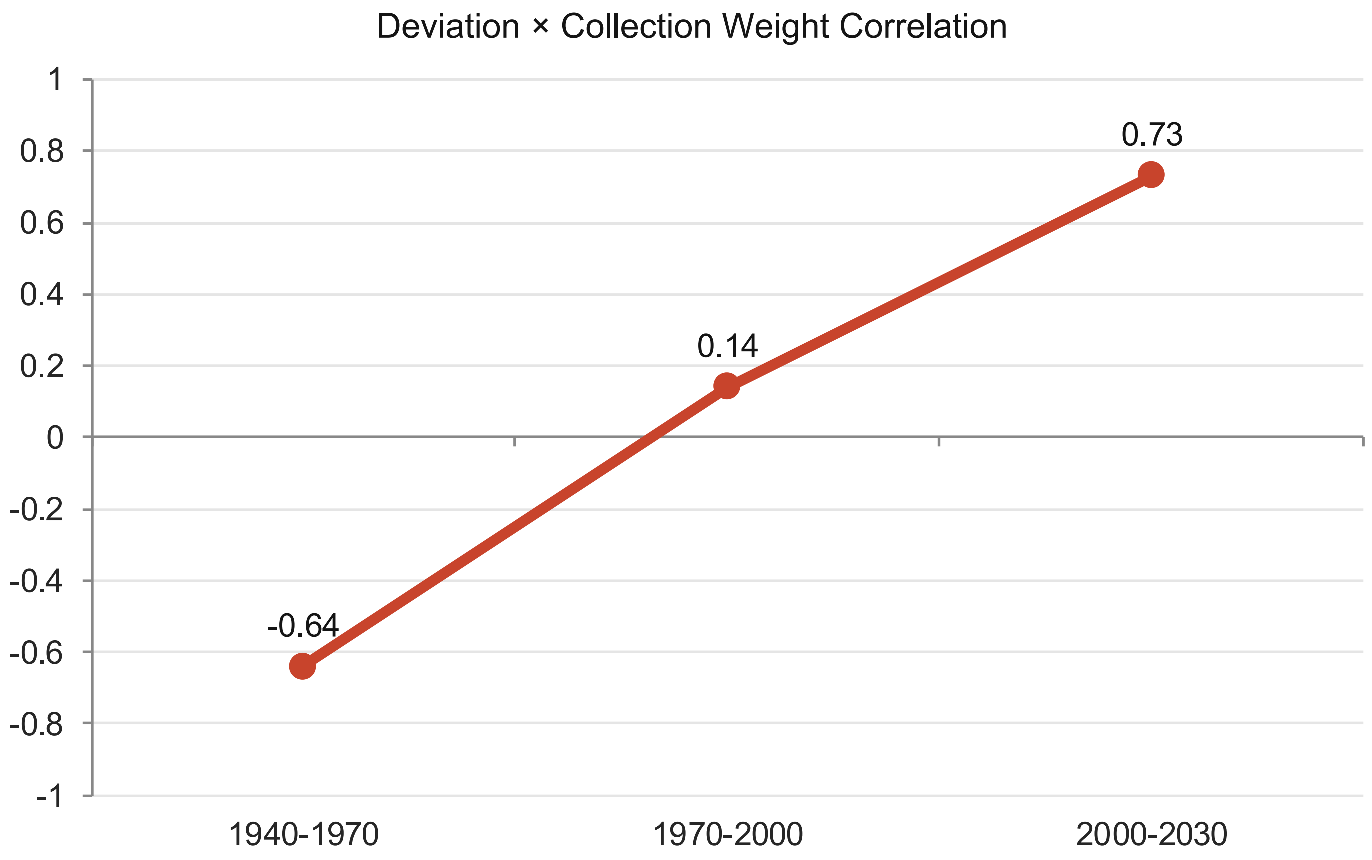


Recognized early: Expressionism, Neues Bauen, modernism — movements MoMA championed during its founding era.

Recognized late: Minimalism, feminist art, abstract art — the main candidates in the collection's catch-up process.

H3 · Art Movement — The Catch-Up Period

“Recognize early / collect heavily” flipped direction over time



1940-1970

r = -0.64 (p=0.015)

Early-recognized movements were collected heavily — MoMA focused on its own contemporaries.

2000-2030

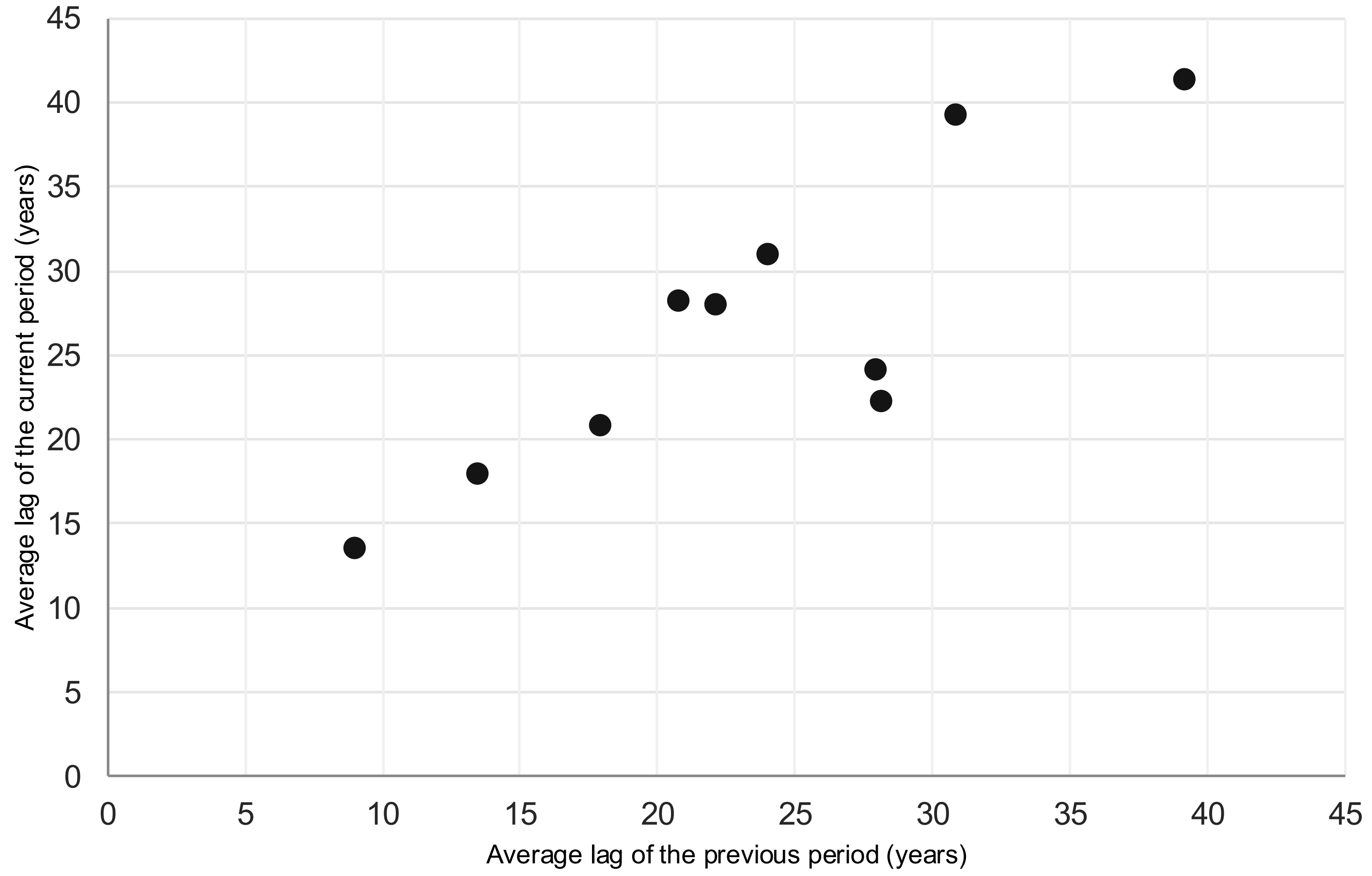
r = +0.73 (p=0.006)

The picture reversed: movements once overlooked (Minimalism, feminist art...) are now gaining weight.

Interpretation: MoMA focused on its contemporaries during its founding era; in later periods, it collects in a way that makes up for past gaps.

H4 · Temporal Consistency

Can one period's lag predict the next — or is the behavior random?



r = 0.855

Autocorrelation coefficient

p = 0.0016 · n = 10

A strong, statistically significant relationship — one period's lag largely predicts the next.

Conclusion

MoMA is not random or unpredictable — it shows consistent, stable institutional behavior. The 2100 projection shows the lag approaching equilibrium at around 45 years.

Bonus Model · Early-Death Artist Projection

How well does the first 8 years of output predict a full career?

$R^2 = 0.705$

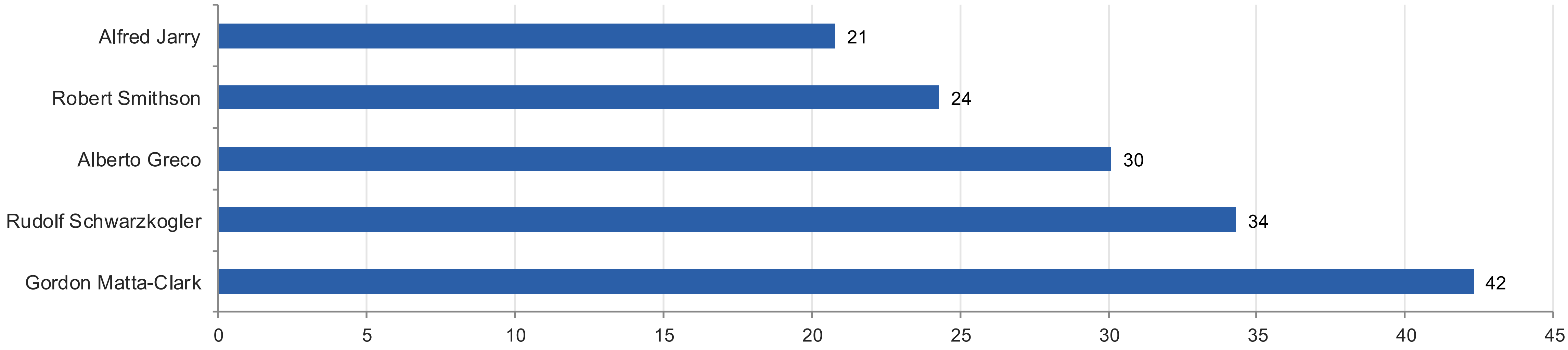
$\log(\text{total}+1) = 1.057 \times \log(\text{first8years}+1) + 0.182$

8-year threshold · log-transformed regression · n=5,656 artists

Validation

Keith Haring: actual 49 → predicted 48.8 (near-perfect)
Jean-Michel Basquiat: actual 12 → predicted 17 (+5 “lost potential”)

Highest “Lost Potential” — among 187 artists who died young



Conclusion

- MoMA's gender lag gap has closed statistically — but numerical parity still needs ~80 more years.
- Geographic inequality narrowed over time but never lost statistical significance — a more persistent pattern.
- MoMA's movement preference reversed over time: it focused on its contemporaries at founding, and is now making up for its past.
- The institution's acquisition behavior isn't random — it shows strong temporal consistency ($r=0.855$).

Detailed statistical methodology, limitations, and methodological notes are in the accompanying report and dashboard.

Thank you — Questions?